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Outdoor comfort in historic spaces under thermoacoustic interactions

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Abstract

Outdoor comfort in historic spaces has received increasing attention under the pressures of urban heat and environmental noise, yet thermoacoustic research has focused mainly on modern urban settings. This study investigated Jinci Temple, a historic site in China's cold region. Field measurements and questionnaire surveys were conducted in four open-space types to examine the relationships among thermal conditions, acoustic conditions, and outdoor comfort. Representative optimization scenarios were simulated using ENVI-met and CadnaA and prioritized through an exploratory assessment informed by Heritage Impact Assessment (HIA). Respondents exhibited relatively high summer heat tolerance. The neutral physiological equivalent temperature (NPET) was 29.4 °C, the neutral PET range was 23.7–35.1 °C, and the upper acceptable PET threshold was 34.3 °C. Acoustic evaluation varied markedly with perceived sound-source type. At a neutral mean acoustic sensation vote, sound pressure levels were 72.2, 68.8, and 60.9 dB for positive, neutral, and negative sound sources, respectively, indicating greater acoustic tolerance under positive sound conditions. Thermoacoustic associations were context-dependent: thermally comfortable conditions were associated with lower sensitivity to SPL variation, whereas moderate acoustic conditions were associated with weaker thermal discomfort. Scenario simulations predicted that reversible shading with a solar transmittance of 0.3 would reduce mean radiant temperature and PET by 12.5 and 5.3 °C, respectively, whereas the predicted sound pressure level reductions for the 3-m and 5-m multilayer green buffers were only 0.2 and 0.4 dB, respectively. HIA identified vegetation shading and greening as priority strategies, water–vegetation synergy as a localized conditional strategy, and visitor noise management as a supplementary measure.

Keywords

historic spaces
outdoor thermal comfort
acoustic comfort
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cold region

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1 Introduction

Rapid urbanization has increased thermal and acoustic pressures in many Chinese cities. These pressures affect outdoor comfort and environmental quality in public spaces (Yin et al. 2022; Ren et al. 2024; Zhan et al. 2024). Taiyuan is a highly urbanized city with a long historical legacy. In 2025, its urbanization rate reached 89.67%, and its urban population reached 4.94 million (Taiyuan Municipal Bureau of Statistics 2025). These conditions make Taiyuan a representative case of intensified urban environmental pressure. Historic spaces are important carriers of urban culture and public activity. They differ from general urban open spaces in spatial form, surface materials, vegetation cover, and visitor-use patterns. Under combined heat and noise stress, it is important to examine the coupled effects of thermal and acoustic environments in historic spaces. Such research can help improve visitor experience, protect public health, and support the sustainable use of cultural heritage.

Early studies on outdoor comfort mainly focused on the effects of individual environmental factors, such as temperature, humidity, or noise, on human perception. However, comfort evaluation in real environments is usually influenced by multiple sensory factors. As multisensory environmental research has developed, thermoacoustic interactions have received increasing attention. Yin et al.

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List of symbols

ACV	acoustic comfort vote	p	p value
AI	acoustic and soundscape improvement potential	PET	physiological equivalent temperature (°C)
ASV	acoustic sensation vote	PPD	predicted percentage of dissatisfied (%)
clo	clothing insulation	RH	relative humidity (%)
ES	entrance square	r_s	Spearman's rank correlation coefficient
F	feasibility	SP	space inside the pavilion
G	global radiation (W/m^2)	SPL	sound pressure level (dB)
HC	heritage compatibility	SVF	sky view factor
LAD	leaf area density (m^2/m^3)	T_a	air temperature (°C)
MACV	mean acoustic comfort vote	T_g	globe temperature (°C)
MASV	mean acoustic sensation vote	T_{mrt}	mean radiant temperature (°C)
mean ± SD	mean ± standard deviation	TCV	thermal comfort vote
met	metabolic equivalent	TI	thermal improvement potential
MTCV	mean thermal comfort vote	TS	tree shade space
MTSV	mean thermal sensation vote	TSV	thermal sensation vote
NPET	neutral physiological equivalent temperature (°C)	V_a	wind speed (m/s)
NPETR	neutral physiological equivalent temperature range (°C)	WS	waterfront space

(2022) used field measurements and questionnaire surveys in campus open spaces in Xi'an and found that both thermal and acoustic environments affected comfort evaluation, with the thermal environment usually playing a dominant role. Studies on campus open spaces in cold regions (Chen et al. 2022), urban squares in the Great Tang All Day Mall in Xi'an (Zhen et al. 2025b), and Fuzhou West Lake Park (Chen et al. 2023b) further showed that sound-source type, sound pressure level, and thermal environmental indicators jointly influence overall outdoor environmental evaluation across different climatic backgrounds and types of urban open spaces. These findings indicate that thermal and acoustic environments do not simply have additive effects but may involve synergistic relationships.

Further studies have shown that thermoacoustic interactions are influenced not only by physical environmental conditions but also by social factors. Wu et al. (2019) found that a comfortable thermal environment improved patients' evaluations of the acoustic environment in hospital environments in heating regions of northern China. Du et al. (2023) developed a prediction model for the overall visual–auditory–thermal comfort of older adults in outdoor residential environments in cold regions. Abbasi et al. (2024) found that thermoacoustic comfort ranges differed by gender and workload in simulated work environments. Jonescu et al. (2025) reported that the combined effects of sound, temperature, and illuminance also affected sleep quality and circadian rhythm in intensive care environments.

In addition, the effects of thermoacoustic interactions have extended from physical comfort evaluation to emotional regulation and psychological restoration. Benfield et al. (2010) found through a national park landscape evaluation experiment that persistent traffic noise reduced spatial comfort and weakened psychological restoration benefits. Herranz-Pascual et al. (2019) reported that restorative emotional effects were significantly associated with environmental comfort, especially acoustic comfort, in several public spaces in Vitoria-Gasteiz, Spain. Bai and Jin (2023) further showed that visual, thermal, and acoustic comfort all contributed to psychological restoration among older adults in severely cold cities. These studies indicate that thermoacoustic interactions have clear group-related differences and psychological effects.

With the development of multisensory environmental research, scholars have begun to examine environmental comfort from a multifactor perspective. Geng et al. (2022), who studied campus open spaces in cold regions, and Xiao et al. (2024), who studied summer campus environments in hot-humid regions, both found significant interactions among thermal, acoustic, and visual environments. Zeng et al. (2024) used multiple linear regression analysis and showed that thermal, acoustic, and visual environments jointly influenced pedestrians' integrated perceptions of temperature, sound pressure level, and lighting conditions in residential areas. Qin et al. (2024) further found that, in cold regions, low sound pressure levels helped mitigate extreme thermal stress, and this effect was moderated by relative humidity. Meanwhile, emerging technologies have

provided new methodological approaches for thermoacoustic research, such as non-contact real-time evaluation based on facial micro-expression recognition (Hu et al. 2022), analysis of cross-modal acoustic–thermal effects in immersive virtual environments (Latini et al. 2023), and environmental perception research combining electroencephalography (EEG) signals and neurobehavioral testing (Sun et al. 2024b). However, existing studies mainly focus on indoor environments or modern urban open spaces. Outdoor historic spaces in real settings remain understudied. The mechanisms of multifactor environmental comfort also require further investigation.

Previous studies have reported marked differences in neutral physiological equivalent temperature (NPET) and neutral physiological equivalent temperature range (NPETR) across cities, climatic backgrounds, and spatial types. These differences indicate that outdoor comfort evaluation is strongly context-dependent (Cohen et al. 2013; Mi et al. 2020; Yin et al. 2021; Liu et al. 2023a; Yuan et al. 2023; Zhen et al. 2025a). In summary, previous research on thermoacoustic associations has focused mainly on modern urban environments, such as campuses, parks, residential areas, and commercial districts. By contrast, historic spaces remain underexplored, despite their distinctive spatial morphology, material characteristics, visitor activities, and conservation constraints. These contextual differences suggest that outdoor comfort in historic spaces may not fully follow the evaluation patterns reported for conventional urban open spaces. In particular, limited evidence is available on two issues: whether thermal and acoustic perceptions differ among open-space types within historic spaces, and whether sound-source type affects thermoacoustic evaluation.

Therefore, this study extends outdoor thermoacoustic comfort research to historic spaces. It focuses on environmental perception under the combined influence of historic spatial morphology, heritage conservation constraints, and sound-source type. Using Jinci Temple as a case study, this study addressed three objectives: (1) to compare thermal and acoustic environmental characteristics and subjective evaluations across representative open-space types within historic spaces; (2) to examine the associations between thermal and acoustic perception, with particular attention to the role of sound-source type; and (3) to quantify selected thermoacoustic environmental improvement scenarios using ENVI-met and CadnaA and subsequently evaluate their environmental benefits, heritage compatibility, and implementation feasibility through an exploratory HIA framework, thereby providing a basis for strategy prioritization. In doing so, the study extends thermoacoustic field research from modern urban spaces to historic spaces. It also suggests that thermoacoustic evaluation may follow context-sensitive patterns under conservation constraints.

2 Methods

2.1 Study area

Taiyuan is the capital of Shanxi Province in North China (37°52'N, 112°33'E). It has a temperate continental monsoon climate. The city is characterized by cold winters, hot summers, short transitional seasons, large diurnal temperature variations, and relatively low but seasonally concentrated precipitation (Peel et al. 2007). Based on monthly meteorological records from May 2015 to May 2025 (http://data.cma.cn), July had the highest monthly mean air temperature in Taiyuan, at 28.4 °C. According to the Chinese standard for the thermal design of civil buildings (MOHURD 2016), Taiyuan belongs to the cold climate zone. Because summer conditions are hot and relatively dry, outdoor environmental comfort warrants particular attention.

Jinci Temple (37°42'23"N, 112°26'57"E) is located in the southwestern suburb of Taiyuan and is a representative historic site in China. Its high historical value, complex spatial structure, abundant vegetation and water features, and intensive visitor activities make it a suitable case for analyzing outdoor thermoacoustic comfort in historic sightseeing spaces.

To cover different combinations of spatial form, surface materials, vegetation and shading conditions, water conditions, and pedestrian activity characteristics, four representative open-space sites were selected in this study (Figure 1): the entrance square (ES), tree shade space (TS), waterfront space (WS), and space inside the pavilion (SP).

The ES site is located in the entrance square at the beginning of the main axis of the building complex. It is mainly characterized by marble hard paving, an open spatial form, and limited tree-canopy shading. Compared with the other vegetated sites, the relatively low mean leaf area density (LAD; 0.16 m^2/m^3) and high sky view factor (SVF; 0.71) indicate sparse vegetation and high sky exposure, respectively, characterizing ES as a typical open hard-paved plaza. The TS site is located in a shaded resting area with surrounding green boundaries. It is mainly influenced by canopy shading from mature broad-leaved trees, such as ancient Chinese scholar trees. The low SVF (0.30) indicates limited sky exposure under the mature canopy, while the mean LAD (0.24 m^2/m^3) indicates a denser canopy structure than at ES, characterizing TS as a semi-shaded tree-canopy space. The WS site is located along the Zhibo Canal. It consists of stone paving, waterfront vegetation comprising willows, other mature trees, shrubs, and grasses, and an adjacent narrow open water surface. Its mean LAD (0.28 m^2/m^3), the highest among the vegetated sites, indicates a relatively dense local vegetation structure. The SVF (0.34) indicates

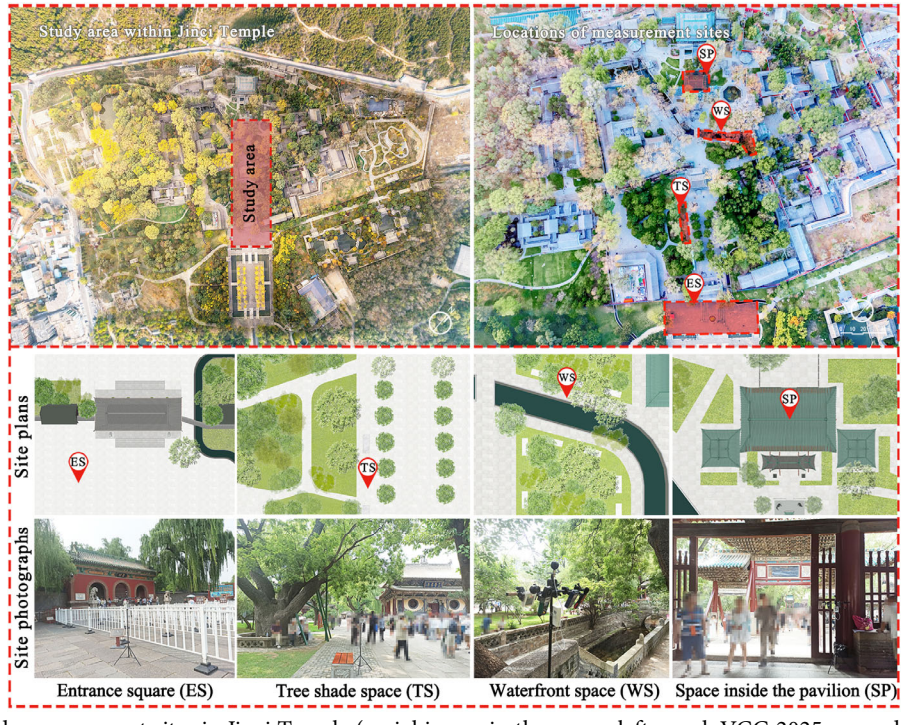


Fig. 1 Study area and measurement sites in Jinci Temple (aerial image in the upper-left panel: VCG 2025; reproduced with permission © Visual China Group)

slightly greater sky exposure than at the TS site. Together, these characteristics define a waterfront resting space influenced by both vegetation and the adjacent water body. The SP site is located inside Xian Hall on the main axis. Shading is primarily provided by the roof structure and architectural enclosure rather than vegetation. Its very low SVF (0.06) indicates a high degree of spatial enclosure and limited sky exposure. Mean LAD was not calculated because no substantial vegetation canopy was present within the measurement area. The site therefore represents a building-enclosed semi-open space. Airflow and solar exposure are important factors affecting thermal comfort in such spaces (Li et al. 2025a).

The main characteristics of each measurement site in terms of surface materials, vegetation structure and spatial shading indicators, water conditions, and pedestrian activity characteristics are shown in Table 1.

2.2 Experimental design

2.2.1 Environmental measurements

Data were collected in July 2025. According to recent

Table 1 Physical characteristics of each measurement site

Space	Spatial form	Surface materials	Vegetation structure and spatial shading indicators	Water conditions	Pedestrian activity characteristics
ES	Open hard-paved plaza	Marble hard paving	Mean LAD = 0.16 m^2/m^3; SVF = 0.71; sparse vegetation and discontinuous tree-canopy shading	No nearby water body	Concentrated pedestrian movement and staying activities
TS	Semi-shaded tree-canopy space	Brick–stone paving with surrounding green boundaries	Mean LAD = 0.24 m^2/m^3; SVF = 0.30; canopy shading provided by mature broad-leaved trees, including ancient Chinese scholar trees	No nearby water body	Dispersed activities, mainly resting
WS	Waterfront resting space	Stone paving combined with waterfront vegetation	Mean LAD = 0.28 m^2/m^3; SVF = 0.34; relatively dense waterfront vegetation comprising mature trees, including willows, shrubs, and grasses	Adjacent to the Zhibo Canal, a narrow open water surface	Dispersed activities with low staying intensity
SP	Building-enclosed semi-open space	Stone hard paving inside the building	Mean LAD: N/A; SVF = 0.06; no substantial vegetation canopy, with shading provided by the roof and architectural enclosure	No nearby water body	Short-term staying with local gathering

meteorological data from Taiyuan, July is a typical high-temperature month with strong heat exposure. Therefore, this study selected this period for field measurements and questionnaire surveys to capture the thermoacoustic environmental characteristics of the study site under typical high-temperature summer visiting conditions. To reduce the influence of abnormal weather on thermoacoustic measurements, surveys were conducted only on sunny or cloudy days; rainy days were excluded. The survey lasted for one week, with measurements conducted daily from 9:00 to 17:00. This period covered the main daytime visiting hours and included both weekdays and weekends. This period captured part of the diurnal variation during high-temperature summer visiting periods. Therefore, the results should be interpreted as representing the thermoacoustic conditions of Jinci Temple under typical high-temperature summer visiting conditions.

Air temperature (T_a), relative humidity (RH), globe temperature (T_g), global radiation (G), wind speed (V_a), and sound pressure level (SPL) were monitored simultaneously at the four measurement sites (Table 2). Globe temperature integrates the effects of air temperature, radiation, and convection (Zhou et al. 2024a). It was used to derive mean radiant temperature (T_{mrt}), which was calculated according to ISO 7726 (ISO 2025a):

$$T_{mrt} = \left[\left(T_g + 273 \right)^4 + \frac{1.10 \times 10^8 V_a^{0.6}}{\varepsilon D^{0.4}} \left(T_g - T_a \right) \right]^{\frac{1}{4}} - 273 \quad (1)$$

where, D denotes the globe diameter (D = 0.05 m in this study), and ε denotes emissivity (ε = 0.95 for the black globe).

Instrument locations were selected in open areas with concentrated visitor activity at each measurement site and were arranged to avoid local influences from vertical interfaces, such as walls and tree trunks, where possible. Because the four spatial types differed considerably in morphology, no fixed uniform distance was set between the instruments and surrounding vertical interfaces. Instead, the instruments were positioned according to field conditions in locations that could represent the environmental characteristics of the main activity areas within each space. For the SP site, shading, radiative reflection, and acoustic reflection caused by the roof, columns, and surrounding building interfaces were regarded as inherent environmental characteristics of this semi-open space rather than measurement errors. Therefore, this study did not separately measure the reflection parameters of building interfaces.

Simultaneously with the thermoacoustic measurements, vertically upward-facing fisheye images were captured at a height of 1.5 m at each measurement point to quantify vegetation structure and sky openness (Konarska et al. 2021). The hemispherical images were processed in CAN-EYE using binary sky–vegetation classification, and the leaf area index (LAI) was estimated from canopy gap fraction and light-transmission characteristics at different zenith angles. The SVF was calculated in RayMan Pro as the weighted proportion of visible sky. SVF ranges from 0 to 1, with higher values indicating greater sky exposure. Mean LAD was calculated by dividing LAI by the effective canopy depth as follows (Weiss et al. 2004):

$$\text{Mean LAD} = \frac{\text{LAI}}{H - H_b} \quad (2)$$

where H is the tree height, H_b is the height of the canopy base, and ($H - H_b$) represents the effective canopy depth. Mean LAD and SVF were used to characterize foliage density and spatial openness, respectively. Mean LAD was used to quantify differences in canopy structure among the measurement points and vegetation scenarios, thereby providing a consistent structural parameter for scenario comparison. Because mean LAD is not a direct acoustic input parameter in CadnaA, it was not converted directly into a vegetation sound-absorption coefficient.

The main areas of Jinci Temple are mostly paved with hard materials, such as marble, brick, and stone, and the surface materials at the measurement sites were generally similar. This study did not separately measure surface albedo or material sound absorption coefficients. The combined effects of surface materials and surrounding interfaces were reflected indirectly through T_g, G, and SPL. Before the study, all instruments were calibrated and mounted on stands at a height of 1.5 m above ground level. Meteorological parameters were recorded at 1-min intervals. Considering the large fluctuations in outdoor SPL, the acoustic environment was represented by the mean SPL recorded during

Table 2 Specifications of the instruments used for environmental measurements

Instrument	Parameter	Range	Accuracy
Taiwan Hengxin AZ87783	Air temperature	0 to 50 °C	±0.6 °C
Heat Index Meter	Relative humidity	0.1% to 99 %	±0.6%
Jianda Renke RS-HQ black globe thermometer	Globe temperature	−40 to 120 °C	±0.2 °C
Xinbao Solar Power Meter SM206	Global radiation	0.1 to 1999.9 W/m^2	0.1 W/m^2
Seta Anemometer	Wind speed	0 to 30 m/s	±0.2 m/s
BSWA 309	Sound pressure level	A LO (low) 0 to 160 dB	0.1 dB

questionnaire completion, and the main on-site sound sources were recorded simultaneously. Abnormal acoustic peaks were included in the descriptive statistics, but were interpreted with caution in the discussion.

2.2.2 *Questionnaire survey*

The questionnaire survey was conducted simultaneously with the field measurements. Given the high mobility of visitors on site, strict probability random sampling based on a complete visitor list was difficult to implement. Therefore, this study adopted an intercept survey method. During fixed survey periods at the four measurement sites, investigators invited visitors passing through or staying near the measurement sites at preset time intervals. Respondents were not selected according to age, gender, visitor identity, or companion characteristics. Environmental parameters were recorded simultaneously while respondents completed the questionnaire. Respondents were mainly engaged in typical low-intensity activities in historic spaces, such as sitting, standing, or slow walking, corresponding to approximately 1.1–1.2 metabolic equivalents (met). All participants were informed of the purpose and procedure of the study and participated voluntarily.

The questionnaire consisted of two parts (see the Appendix for the complete questionnaire in the Electronic Supplementary Material of the online version of this paper). The first part collected basic respondent information, including gender, age, height, weight, and visitor status. The second part recorded subjective evaluations of the thermal and acoustic environments, including thermal sensation vote (TSV), thermal comfort vote (TCV), thermal acceptability, thermal preference, acoustic sensation vote (ASV), acoustic comfort vote (ACV), acoustic preference, and overall comfort. Following ASHRAE thermal comfort evaluation methods, seven-point and three-point Likert scales were used (Table 3). The main sound sources heard by respondents were also recorded and classified, including broadcast sounds, tour guide sounds, machine noise, music, vehicle sounds, commercial sounds, mobile phone sounds, visitor voices, tower bell sounds, wind sounds, flowing water sounds, bird songs, and other animal sounds. In mixed acoustic environments, classification was based on the dominant perceived sound source at the time of the survey.

A total of 522 questionnaires were collected in this study. After excluding questionnaires with incomplete responses or logical inconsistencies, 504 valid questionnaires remained. *G**Power was used to conduct an a priori statistical power analysis to evaluate the adequacy of the sample size. The results showed that, with a small effect size of $\rho = 0.16$, a significance level of $\alpha = 0.05$, and a statistical power of $1 - \beta = 0.95$, the minimum required sample size was approximately 500. Therefore, the 504 valid questionnaires obtained in this study met the statistical power requirement.

2.3 Data analysis and simulation evaluation

Excel 2021, IBM SPSS Statistics 27.0, RayMan Pro 2016, OriginPro 2025, ENVI-met 5.7.1, and CadnaA 2023 were used for data processing, statistical analysis, visualization, and optimization-scenario simulations. Specifically, Excel 2021 was used for data cleaning and descriptive statistics. RayMan Pro 2016 was used to calculate physiological equivalent temperature (PET) based on the measured meteorological variables and respondents' personal parameters. IBM SPSS Statistics 27.0 was used for nonparametric correlation analysis and grouped comparative analysis. OriginPro 2025 was used for curve fitting and figure preparation. ENVI-met 5.7.1 and CadnaA 2023 were used to simulate selected thermal and acoustic environmental optimization scenarios, respectively. Figure 2 illustrates the overall research workflow, including field measurements, questionnaire surveys, data analysis, and simulation-based evaluation.

The study first used descriptive statistics to compare thermoacoustic environmental parameters and subjective evaluations across different open-space types within the historic site. Because TSV and other subjective evaluation variables are ordinal and some variables did not meet the normality assumption, Spearman correlation analysis was then used to examine bivariate relationships between TSV and measured meteorological parameters. This analysis was

Table 3 Scales used for subjective evaluation

TSV	ASV	TCV, ACV	Preferences for other environmental factors
−3, cold	−3, very loud	−3, very uncomfortable	—
−2, cool	−2, loud	−2, uncomfortable	—
−1, slightly cool	−1, slightly loud	−1, slightly uncomfortable	−1, decrease/weaker
0, neutral	0, moderate	0, neutral	0, unchanged
+1, slightly warm	+1, slightly quiet	+1, slightly comfortable	+1, increase/stronger
+2, warm	+2, quiet	+2, comfortable	—
+3, hot	+3, very quiet	+3, very comfortable	—

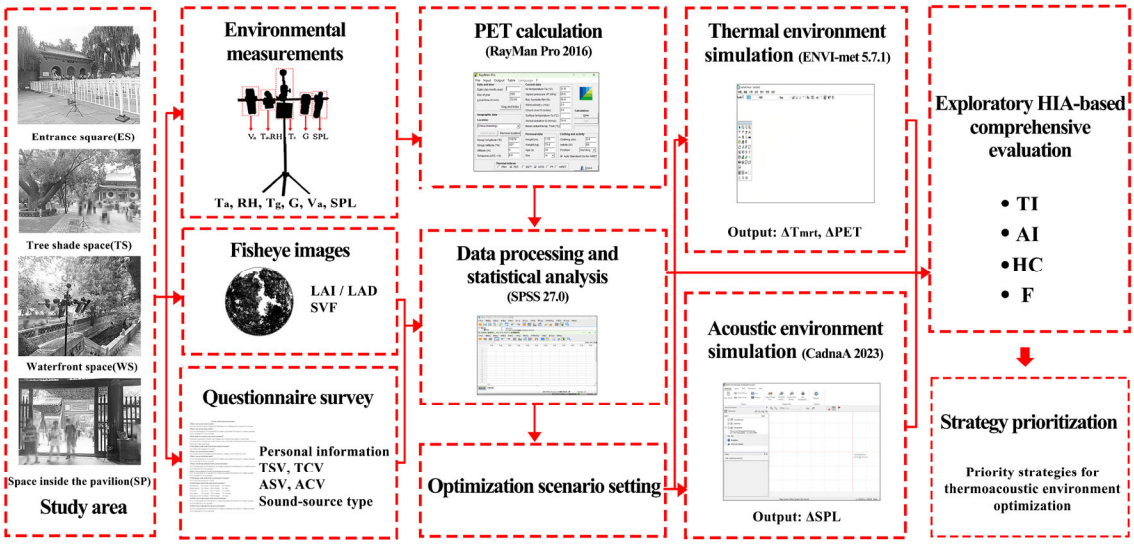


Fig. 2 Research workflow for the field survey, data analysis, and simulation-based evaluation

used to identify preliminary associations between thermal sensation and individual meteorological parameters. On this basis, grouped fitting, threshold analysis, and stratified comparisons were further used to examine whether the relationships among PET, SPL, thermal evaluation, and acoustic evaluation varied with spatial type, thermoacoustic perception category, and sound-source type. Based on the field measurements, subjective evaluations, and spatial characteristics of the four sites, selected optimization scenarios were developed and quantitatively compared using ENVI-met and CadnaA. Following the simulation-based comparison, an exploratory HIA framework was applied to prioritize the strategies across four dimensions: thermal improvement potential, acoustic environment and soundscape improvement potential, heritage compatibility, and implementation feasibility.

3 Results

3.1 Descriptive results

3.1.1 Respondent characteristics

A total of 522 questionnaires were collected, of which 504 were valid. Invalid questionnaires were identified based on incomplete responses, logical inconsistencies, overly uniform answers, or evidence of insufficient response effort. The questionnaires were checked on site to improve completeness and consistency and to reduce invalid responses. Among the valid questionnaires, 129 were collected at the ES site, 132 at the TS site, 129 at the SP site, and 114 at the WS site (Table 4). The gender distribution was relatively balanced, with 50.4% female and 49.6% male. The 26–40 age group accounted for the largest proportion, at 48.8%. Out-of-town visitors (defined as Chinese visitors from outside Taiyuan) accounted for 51.2% of the sample, slightly exceeding the proportion of local residents (48.8%). Overall, the sample was relatively balanced in terms of measurement site, gender, and visitor status. This sample therefore provides a basis for comparative analysis among different spatial types during summer daytime conditions.

Given that the climatic background of an individual's long-term place of residence may be associated with thermal perception (Su et al. 2024), respondents were preliminarily classified in the questionnaire as originating from northern or southern China. Most non-local visitors came from northern regions with climatic conditions similar to those of the study area, indicating only limited differences in

Table 4 Demographic characteristics of participants

Variable	Category/statistic	Value
Research site	Total	504
	ES	129
	TS	132
	SP	129
	WS	114
Biological sex	Male	250
	Female	254
Status	Local residents	246
	Out-of-town visitors	258
Age (year)	Mean ± SD	33.4 ± 13.7
Height (m)	Mean ± SD	1.7 ± 0.1
Weight (kg)	Mean ± SD	64.2 ± 12.2
Clothing insulation (clo)	Mean ± SD	0.4 ± 0.1
Metabolic rate (met)	Mean ± SD	1.1 ± 0.1

climatic exposure. Therefore, subsequent analyses did not distinguish between local residents and non-local visitors. All respondents were treated as a single group.

3.1.2 Environmental conditions at the four measurement sites

The thermoacoustic environmental parameters differed markedly among the four measurement sites (Table 5). The ES site was typical of an unshaded open space. It had the highest mean G (419.4 W/m^2) and mean T_a (36.7 °C), the largest air temperature range (29.3–40.1 °C), and the lowest mean RH (30.3%), indicating a high level of heat exposure. Its mean SPL was also relatively high (71.5 dB), which may be related to concentrated pedestrian flow and frequent activities in the entrance area. The parameters at the TS site were generally at intermediate levels, with mean G, T_a, RH, and SPL of 137.3 W/m^2, 34.5 °C, 36.4%, and 67.8 dB, respectively. This indicates that tree shade reduced global radiation and heat exposure to some extent. The SP site was located inside a semi-open pavilion. It had the lowest mean G (3.5 W/m^2) and relatively small fluctuations, reflecting the strong effect of building shading on global radiation. It also had the lowest mean V_a (0.2 m/s), indicating that the semi-enclosed space may have restricted air movement. Meanwhile, the SP site had the highest mean SPL (83.1 dB) and a relatively large standard deviation, indicating marked acoustic fluctuations. This may be related to short-term visitor gatherings and transient sound events within the semi-enclosed space. The WS site had the highest mean RH (44.1%) and the lowest mean SPL (61.3 dB). It also had relatively low mean G (41.6 W/m^2) and moderate V_a (0.6 m/s). This suggests that the combined effects of vegetation shading and water bodies helped reduce heat exposure and improve the acoustic environment.

3.1.3 Thermal sensation vote and thermal comfort vote (TSV and TCV)

Figure 3(a) shows that overall TSV responses were predominantly "neutral" (TSV = 0), accounting for 62.7% of all responses. "Slightly warm" (TSV = +1) and "warm" (TSV = +2) accounted for 16.1% and 15.5%, respectively, while "hot" (TSV = +3) accounted for 4.4%. At the site level, the ES site had the highest proportion of "warm" votes (TSV = +2), at 39.4%, markedly higher than the other sites. The distributions at the TS and SP sites were similar, whereas respondents at the WS site tended to perceive the environment as cooler, likely owing to the canal-side setting and surrounding vegetation. Figure 3(b) indicates that, for overall TCV, 75.2% of respondents rated the environment as moderate or above (TCV ≥ 0). Among the four sites, the ES site showed the poorest thermal comfort, with 68.2% of respondents reporting discomfort (TCV ≤ −1), consistent with its warmer thermal conditions. In contrast, the TS site showed the highest thermal comfort, with 91.7% of respondents assigning ratings of moderate or above (TCV ≥ 0).

Table 5 Environmental parameters at each measurement site

Space		ES	TS	SP	WS
Air temperature (°C)	Max	40.1	40.1	36.6	35.6
	Min	29.3	29.9	30.2	27.1
	Mean ± SD	36.7 ± 3.3	34.5 ± 1.0	34.3 ± 3.5	33.7 ± 2.3
Relative humidity (%)	Max	41.6	60.0	59.5	57.9
	Min	23.5	28.2	32.2	34.5
	Mean ± SD	30.3 ± 3.5	36.4 ± 2.4	39.0 ± 3.2	44.1 ± 10.1
Wind speed (m/s)	Max	3.5	0.9	0.6	0.9
	Min	0.0	0.0	0.0	0.0
	Mean ± SD	0.7 ± 0.3	0.5 ± 0.1	0.2 ± 0.1	0.6 ± 0.3
Globe temperature (°C)	Max	43.3	38.1	36.5	36.5
	Min	30.1	29.8	26.2	31.5
	Mean ± SD	38.5 ± 1.9	33.9 ± 0.1	33.8 ± 2.6	33.7 ± 0.7
Global radiation (W/m^2)	Max	765.3	432.2	6.3	116.8
	Min	36.4	63.5	2.6	29.6
	Mean ± SD	419.4 ± 381.3	137.3 ± 21.2	3.5 ± 0.2	41.6 ± 29.8
Sound pressure level (dB)	Max	87.6	79.3	150.5	69.8
	Min	54.5	53.2	56.3	18.0
	Mean ± SD	71.5 ± 8.9	67.8 ± 1.4	83.1 ± 22.7	61.3 ± 1.1

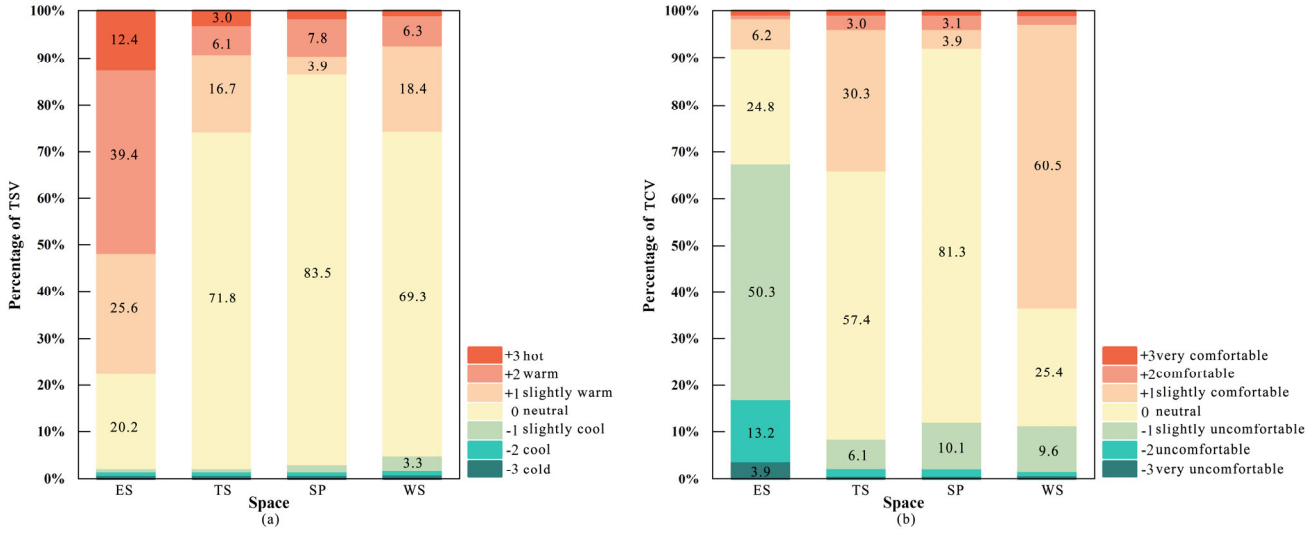


Fig. 3 TSV and TCV for different spaces: (a) TSV; (b) TCV

3.1.4 Acoustic sensation vote and acoustic comfort vote (ASV and ACV)

To simplify acoustic-environment classification and analyze the effects of different sound sources on comfort, this study referred to the classification approach of Chen et al. (2023a), which considers perceived sound-source attributes, contextual meaning, and disturbance in urban park thermoacoustic research. Based on this approach and the field questionnaire results, the dominant sound sources perceived by respondents were classified into three types: positive, neutral, and negative sound sources. Positive sound sources mainly included natural and culturally distinctive sounds (wind sounds, bird songs, flowing water sounds, tower bell sounds, and other animal sounds). Neutral sound sources mainly represented sounds associated with routine human activities (tour guide sounds, broadcast sounds, music, commercial sounds, mobile phone sounds, and visitor voices). Negative sound sources mainly referred to machine noise and vehicle sounds.

To strengthen the objectivity of this classification, the grouped sound-source categories were further compared using measured SPL characteristics, including mean level and short-term variability during the survey period. The results showed that positive sound sources were generally associated with lower SPLs and smoother temporal variation. They were mainly found in shaded green spaces, such as TS and WS. Neutral sound sources were typically associated with moderate SPLs and intermittent fluctuations and were commonly found in areas with concentrated human activity (ES and SP). Negative sound sources tended to occur at relatively higher SPLs and with stronger short-term variability, especially in the open plaza (ES). Therefore, the classification reflects not only perceived acoustic quality and contextual meaning but also differences in basic objective acoustic characteristics. This perception-oriented classification is mainly based on the dominant sound source perceived by respondents and may be affected by individual preferences, cultural background, previous experience, and momentary attention (Sun et al. 2024a).

Figure 4(a) shows that, in terms of overall ASV, 66.3% of respondents rated the acoustic environment as moderate or above (ASV ≥ 0), indicating a generally favorable acoustic environment. However, owing to the characteristics of the open plaza and large transient pedestrian flows, the ES and SP sites received lower ratings than the other sites, suggesting relatively poor acoustic environmental quality. Figure 4(b) indicates that, for overall ACV, 75.2% of respondents assigned ratings of moderate or above (ACV ≥ 0). Among the four sites, the WS site exhibited the highest acoustic comfort, with 97.3% of respondents reporting ratings of moderate or above. In contrast, the proportions of respondents reporting low acoustic comfort (ACV < 0) were relatively high at the ES site (37.2%) and the SP site (54.2%). This pattern is consistent with the ASV results shown in Figure 4(a). As shown in Figures 4(c) and (d), the proportion of positive acoustic comfort votes decreased as perceived sound-source type shifted from positive to negative. In environments dominated by positive sound sources, the proportion of respondents reporting acoustic comfort (ACV > 0) reached 37.5%, which was clearly higher than under the other sound-source conditions, while only 13.1% of respondents reported discomfort (ACV < 0), suggesting that positive sound sources were associated with substantially better acoustic comfort.

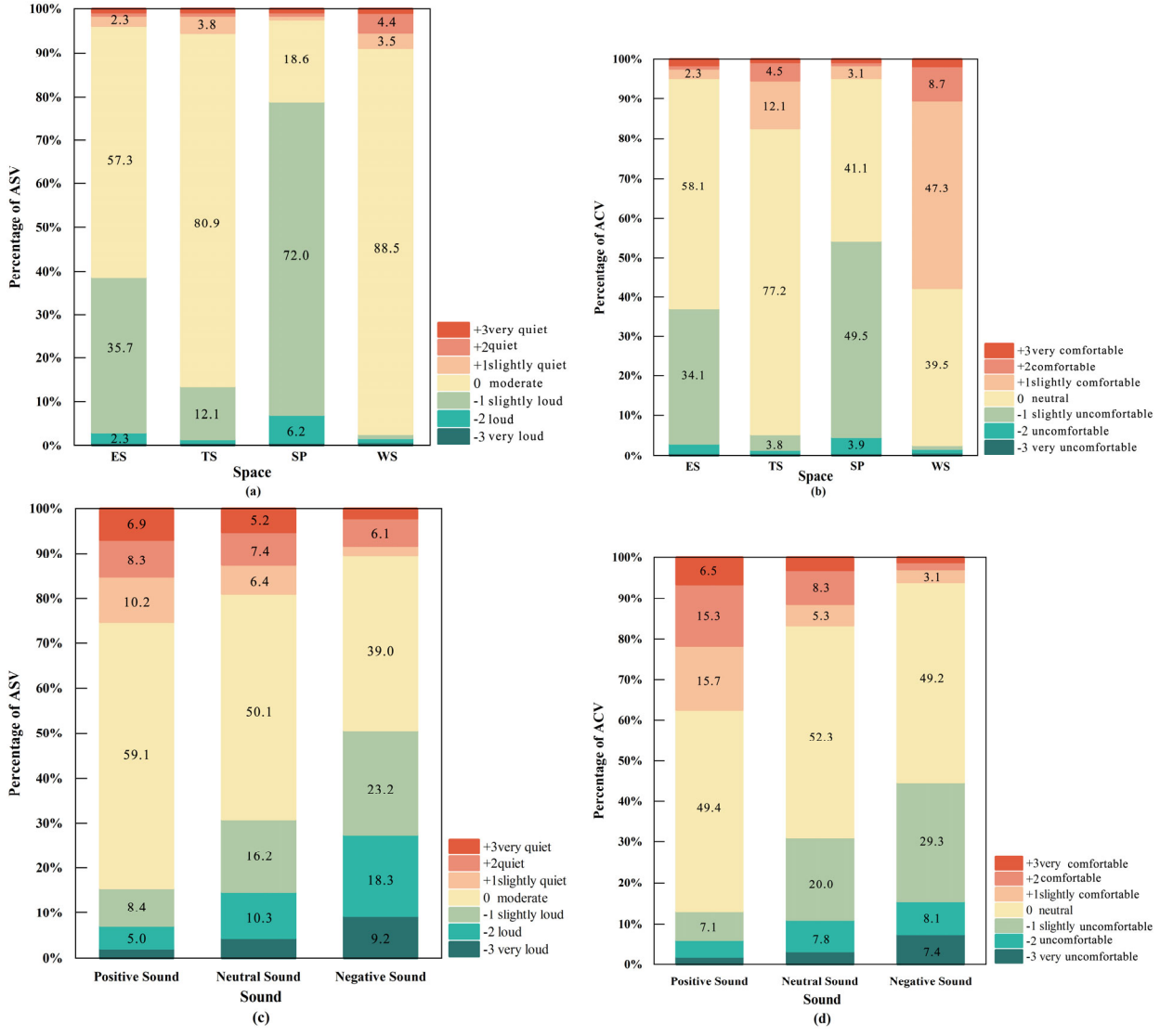


Fig. 4 ASV and ACV under different spaces and sound-source types: (a) ASV for different spaces; (b) ACV for different spaces; (c) ASV for different sound-source types; (d) ACV for different sound-source types

3.2 Correlation analysis

3.2.1 Thermal environment

3.2.1.1 Relationship between thermal sensation and meteorological parameters

Meteorological variables such as air temperature, humidity, wind speed, and radiation may be associated with subjective thermal sensation in outdoor environments (Li et al. 2020). To examine their associations with TSV under different spatial conditions, Spearman correlation analysis was performed separately for each site (Table 6). Spearman's rho (r_s) was used to describe the strength and direction of association, and the corresponding p values were used to assess statistical significance.

The results indicate that T_a was significantly positively correlated with TSV at the ES, TS, and WS sites ($p < 0.05$), suggesting that T_a was an important factor associated with thermal sensation. RH was also significantly correlated with TSV at the ES, TS, and SP sites ($p < 0.05$), further indicating its relevance. In contrast, wind speed showed weak associations with TSV, with generally low correlation coefficients that did not reach statistical significance. At the SP site, T_a and G were weakly correlated with thermal sensation under shaded conditions.

3.2.1.2 Thermal sensation vote and thermal comfort vote

For each TSV category, the corresponding TCV values were averaged to obtain the mean thermal comfort vote (MTCV). Figure 5 presents a quadratic fit of these values, describing the relationship between thermal sensation and thermal comfort during the survey period. The results indicate

Table 6 Spearman correlations between TSV and meteorological parameters

Space	Air temperature		Relative humidity		Wind speed		Global radiation	
	p	r_s	*p*	r_s	*p*	r_s	*p*	r_s
ES	**<0.001****	0.504	**<0.001****	0.619	0.650	−0.041	**<0.001****	0.630
TS	**<0.001****	0.632	**<0.001****	0.589	0.991	0.001	0.229	0.408
SP	0.918	−0.009	**<0.001****	0.514	0.870	0.015	0.813	0.021
WS	**<0.001****	0.730	0.679	0.040	0.834	0.020	**0.003****	0.682

** Significant at $p < 0.01$ (two-tailed); *Significant at $p < 0.05$ (two-tailed).

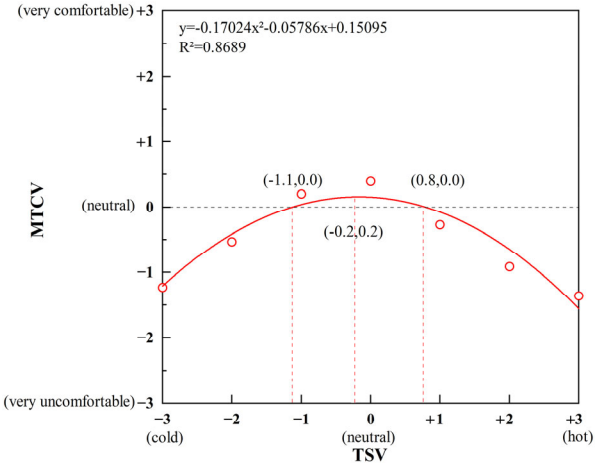


Fig. 5 Relationship between TSV and MTCV

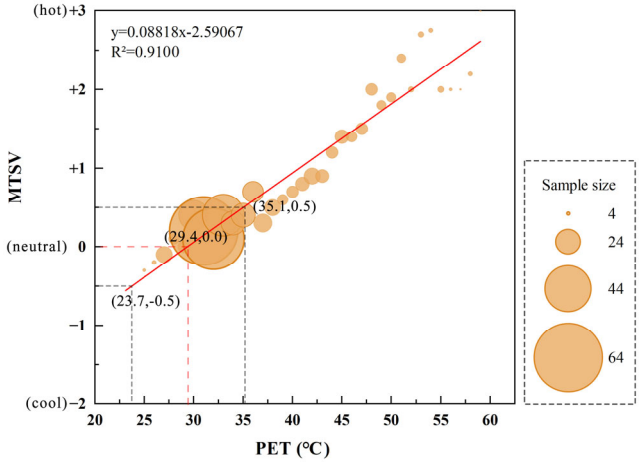


Fig. 6 Relationship between PET and MTSV

that the fitted thermal comfort reached its maximum (MTCV = +0.2) when TSV was approximately −0.2, suggesting that respondents tended to prefer slightly cool to neutral thermal conditions. When −1.1 ≤ TSV ≤ +0.8, the thermal environment was generally perceived as thermally comfortable (TCV ≥ 0).

3.2.1.3 Neutral PET and neutral PET range (NPET and NPETR)

In outdoor thermal comfort studies, neutral temperature is commonly defined as the thermal state in which people feel neither cold nor hot (ISO 2025b). In this study, respondents' basic information, including gender, age, height, weight, clothing insulation, activity state, and metabolic rate, together with meteorological data such as T_a, RH, V_a, and *G*, were entered into the RayMan model to calculate PET. Based on TSV, the weighted MTSV within each PET interval was calculated (Figure 6). The PET corresponding to TSV = 0 was defined as the NPET. The NPET during the survey period was 29.4 °C. The temperature range corresponding to −0.5 ≤ MTSV ≤ +0.5 was defined as the NPETR (Cheung and Jim 2017). As shown in Figure 6, the NPETR during the survey period was 23.7–35.1 °C. Compared with the study conducted at the Xi'an City Wall (Tian et al. 2022) (NPET = 18.6 °C; NPETR = 15.3–20.3 °C), the present study yielded a higher NPET and a wider NPETR, which may reflect differences in climatic background, spatial characteristics, and relatively high thermal tolerance among the surveyed population.

3.2.1.4 Thermal acceptability

According to ASHRAE (2023) Standard 55, thermal acceptability is commonly evaluated using the proportion of respondents who consider the thermal environment acceptable, with 80% acceptability adopted under normal criteria and 90% under more stringent criteria. In this study, the 80% acceptability criterion was adopted, and the thermal unacceptability rate for each 1 °C PET interval was fitted (Figure 7). Based on this criterion, the upper PET threshold for thermal acceptability was estimated as 34.3 °C. This acceptability threshold was substantially higher than the NPET (29.4 °C), suggesting relatively high tolerance to warm conditions during the survey period.

3.2.1.5 Observed thermal dissatisfaction rate

In this study, −1 ≤ TSV ≤ +1 was defined as the thermally comfortable range, whereas −3 ≤ TSV ≤ −2 and +2 ≤ TSV ≤ +3 were defined as thermally uncomfortable ranges (Faraji et al. 2022). By calculating the thermal dissatisfaction rate for each 1 °C PET interval and fitting it to the corresponding MTSV values, an empirical model of thermal dissatisfaction applicable to Taiyuan was established (Figure 8).

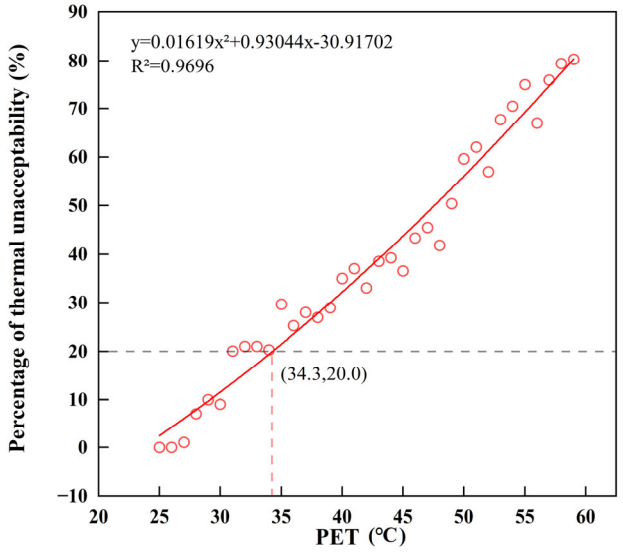


Fig. 7 Relationship between thermal unacceptability rate and PET

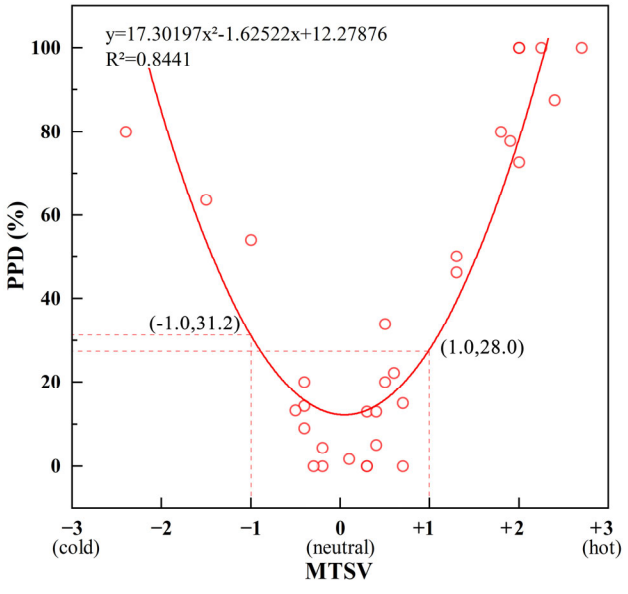


Fig. 8 Relationship between MTSV and thermal dissatisfaction rate

3.2.1.6 *Thermal preference*

Preferred temperature is defined as the ideal thermal condition desired by individuals (Zhang and Lin 2020). Based on the thermal preference votes within each PET interval, Probit analysis identified a summer preferred PET of 34.1 °C (Figure 9), which was 4.7 °C higher than the NPET (29.4 °C). This result suggests that, under the survey conditions, respondents did not necessarily prefer thermally neutral environments. Instead, some respondents may have regarded a slightly warm environment as acceptable or even more preferable.

3.2.2 *Acoustic sensation vote and acoustic comfort vote*

To investigate the relationship between acoustic sensation and acoustic comfort, ACV values were averaged within each ASV category to obtain the MACV, and linear regression was then performed to examine its relationship with MASV (Figure 10(a)). The results showed that acoustic comfort declined as acoustic sensation became more negative. When MACV = 0, ASV was −0.3, indicating that respondents still rated the acoustic environment as acceptable under slightly noisy conditions.

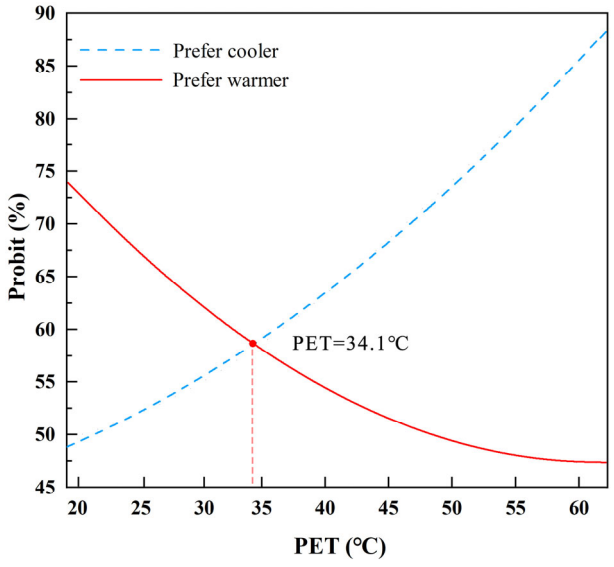


Fig. 9 Preferred PET

The relationship between ASV and ACV was further examined across the three sound-source types (Figure 10(b)). Linear regression showed that the association between acoustic sensation and acoustic comfort was strongest under negative sound-source conditions and weakest under positive sound-source conditions, as indicated by the steepest and shallowest slopes, respectively. At the same acoustic sensation level, acoustic comfort was the lowest for negative sound sources and the highest for positive sound sources.

Further fitting of MASV and MACV across SPL intervals showed that MASV decreased with increasing SPL (Figure 10(c)), and the SPL corresponding to MASV = 0 varied by sound-source type: it was the highest for positive sound sources (72.2 dB), followed by neutral sound sources (68.8 dB), and the lowest for negative sound sources (60.9 dB). MACV also decreased with increasing SPL across the three sound-source types (Figure 10(d)). When MACV = 0, the corresponding SPLs for positive, neutral, and negative sound sources were 76.2 dB, 72.4 dB, and 66.2 dB, respectively. These results indicate that respondents showed the highest acoustic tolerance under positive sound-source conditions and the lowest acoustic tolerance under negative sound-source conditions, suggesting that perceived sound-source type was closely associated with acoustic comfort.

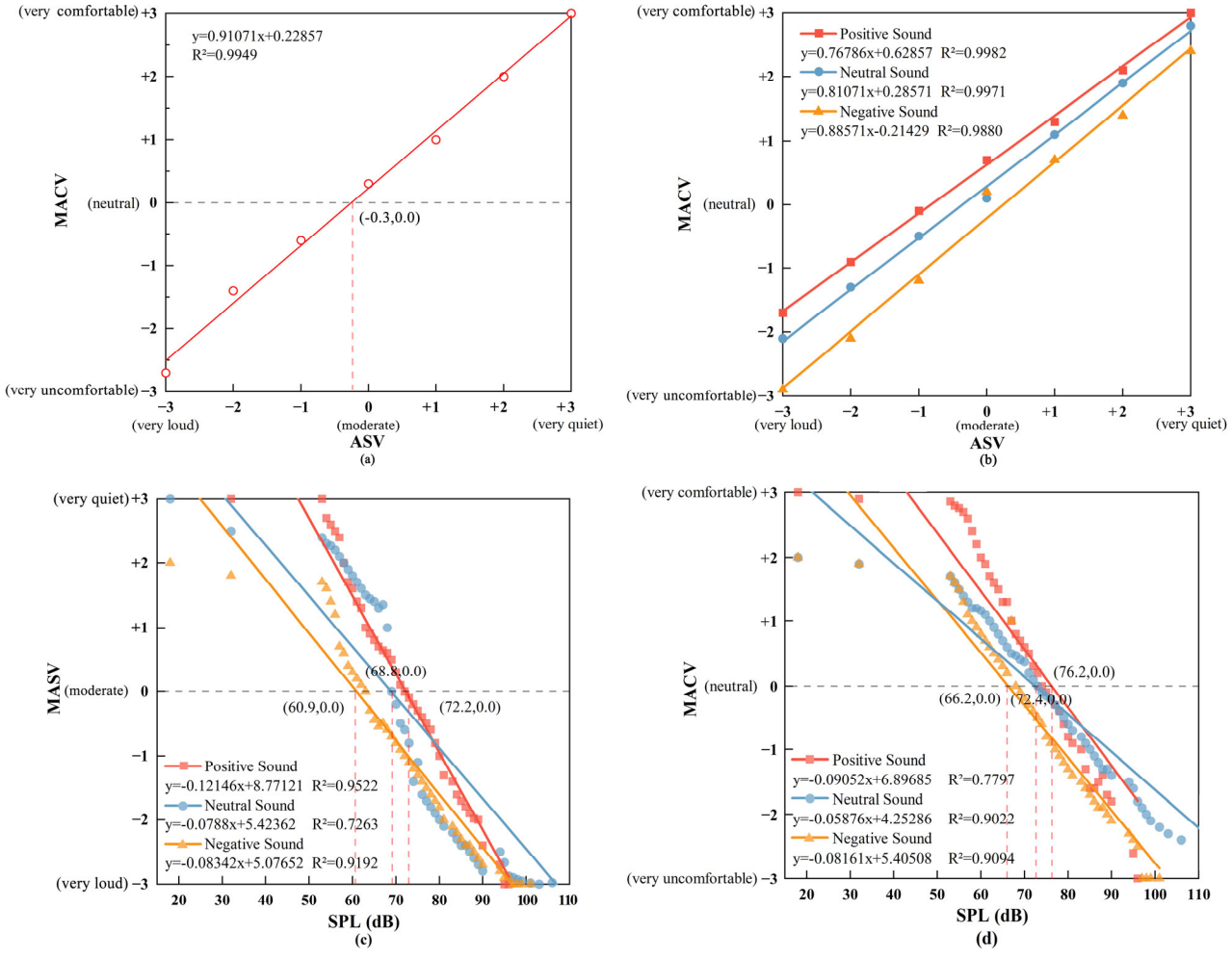


Fig. 10 Relationship between acoustic sensation and acoustic comfort: (a) ASV and MACV; (b) ASV and MACV for different sound-source types; (c) SPL and MASV for different sound-source types; (d) SPL and MACV for different sound-source types

3.3 Effects of thermoacoustic associations on human comfort

To analyze associations between thermoacoustic conditions and subjective comfort evaluation, thermal and acoustic sensation votes were grouped into three levels. Thermal sensation votes were classified as cool (TSV = −3, −2), moderate (TSV = −1, 0, +1), and warm (TSV = +2, +3). Acoustic sensation votes were classified as noisy (ASV = −3, −2), moderate (ASV = −1, 0, +1), and quiet (ASV = +2, +3). Similarly, thermal comfort votes and acoustic comfort votes were categorized as uncomfortable (TCV/ACV = −3, −2), moderate (TCV/ACV = −1, 0, +1), and comfortable (TCV/ACV = +2, +3). This grouping allowed variation patterns in thermal and acoustic evaluation to be analyzed under different conditions.

3.3.1 *Effects of thermal sensation on acoustic evaluation*

The results indicate that SPL was associated with acoustic sensation and that the strength of this association varied across thermal sensation categories. Under cool (TSV = −3, −2) and warm (TSV = +2, +3) conditions, the fitted slopes relating SPL to acoustic sensation ($|k|$ = 0.03474 and 0.02682, respectively) were steeper than the slope observed under moderate thermal sensation ($|k|$ = 0.00816), suggesting that respondents were less sensitive to SPL variation under moderate thermal sensation conditions (Figure 11(a)).

Under different thermal sensation conditions, the fitted relationships between SPL and acoustic comfort were broadly similar ($k \approx -0.043$ to -0.033) (Figure 11(b)). However, the overall level of acoustic comfort still differed across thermal states: acoustic comfort was the highest under cool conditions, followed by moderate conditions, and the lowest under warm conditions. This suggests that warmer thermal conditions were generally associated with less favorable acoustic evaluation.

3.3.2 *Effects of thermal comfort on acoustic evaluation*

Thermal comfort was also associated with acoustic evaluation. The absolute slope between SPL and acoustic sensation

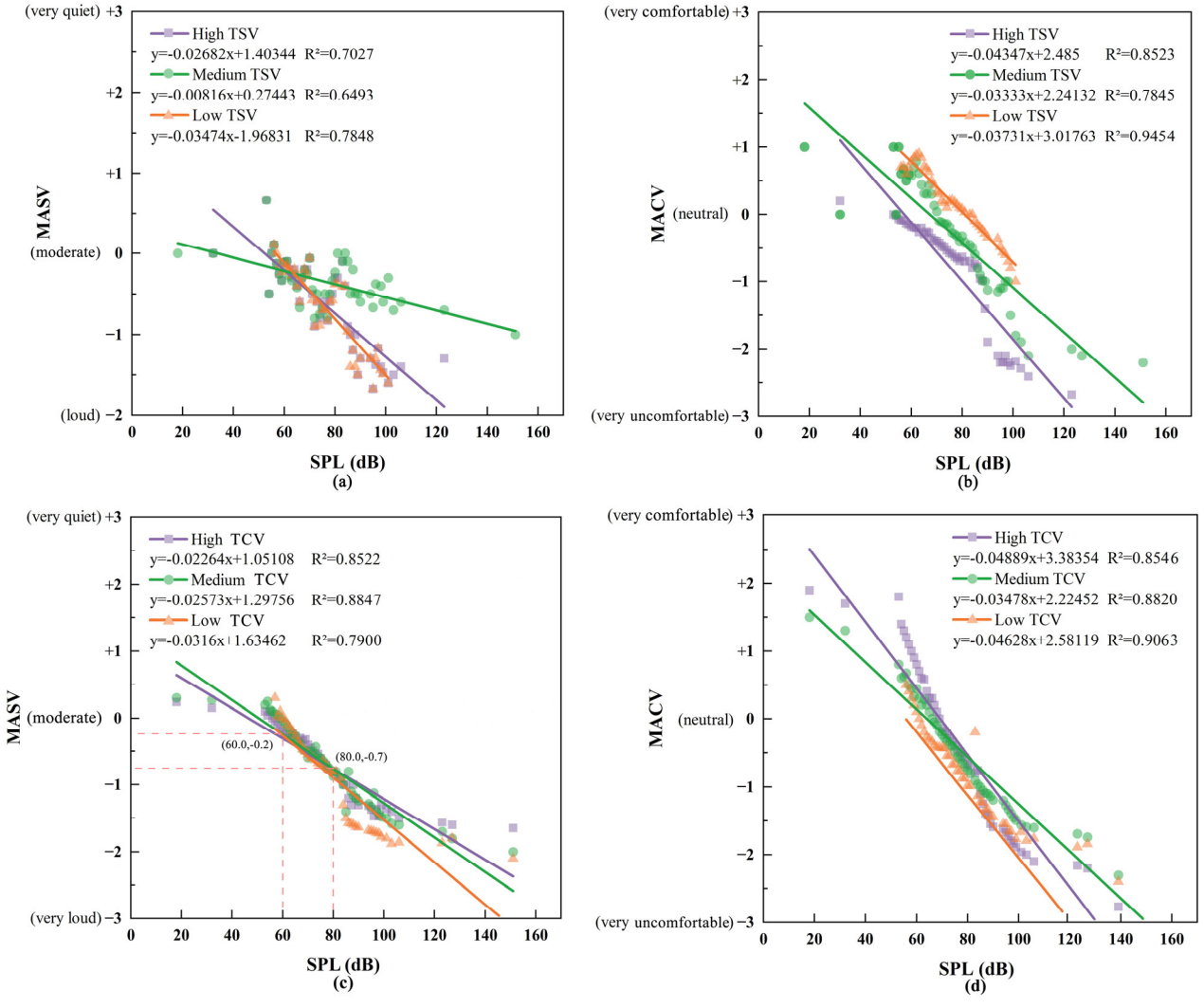


Fig. 11 Relationship between SPL and acoustic evaluation under different thermal conditions: (a) SPL and MASV for different TSV; (b) SPL and MACV for different TSV; (c) SPL and MASV for different TCV; (d) SPL and MACV for different TCV

was smallest under thermally comfortable conditions (TCV = +2, +3; $|k|$ = 0.02264) and largest under thermally uncomfortable conditions (TCV = −3, −2; $|k|$ = 0.0316) (Figure 11(c)). Within the common SPL range of 60–80 dB, MASV was concentrated between −0.2 and −0.7, indicating that the between-group differences existed but were moderate in magnitude. Overall, the results indicate that thermally comfortable conditions were associated with greater tolerance at a given SPL range.

From acoustic comfort (Figure 11(d)), the fitted slope under moderate thermal comfort conditions ($|k|$ = 0.03478) was smaller than those under thermally comfortable ($|k|$ = 0.04889) and thermally uncomfortable ($|k|$ = 0.04628) conditions. Within the measured SPL range, acoustic comfort tended to improve with increasing thermal comfort, which is broadly consistent with previous studies (Lin et al. 2025). In practical terms, this result suggests that improving thermal conditions in crowded outdoor spaces may also contribute to more favorable acoustic evaluation.

3.3.3 *Effects of acoustic sensation on thermal evaluation*

The results indicate that acoustic conditions were associated with thermal sensitivity. The fitted slope relating thermal sensation to PET was the highest in quiet environments (k = 0.10799), followed by moderate acoustic environments (k = 0.09615), and the lowest in noisy environments (k = 0.09022), suggesting that respondents in quieter settings appeared more sensitive to PET variation (Figure 12(a)). However, as PET increased, the differences among the three acoustic types gradually narrowed.

Figure 12(b) shows that the differences in thermal comfort among the three acoustic environments became relatively small under higher PET conditions. The optimal PET corresponding to the fitted curve vertex was 35.3 °C

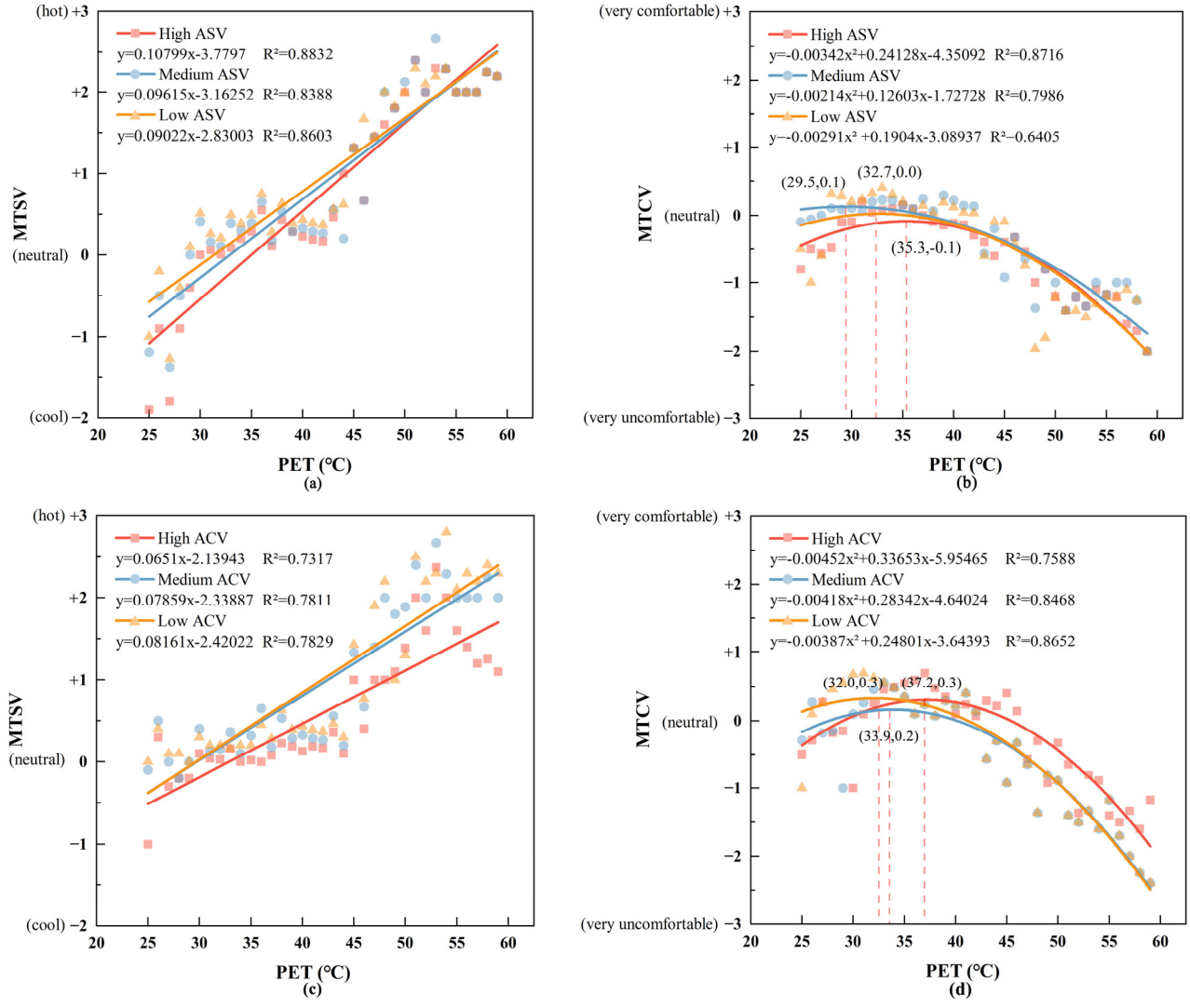


Fig. 12 Relationship between PET and thermal evaluation under different acoustic conditions: (a) PET and MTSV for different ASV; (b) PET and MTCV for different ASV; (c) PET and MTSV for different ACV; (d) PET and MTCV for different ACV

for quiet environments, 29.5 °C for moderate acoustic environments, and 32.7 °C for noisy environments. At these respective optima, thermal comfort was the highest in the moderate acoustic environment (TCV = +0.1), followed by the noisy environment (TCV = 0.0), and the lowest in the quiet environment (TCV = −0.1). These patterns suggest that moderate acoustic conditions were associated with slightly more favorable thermal comfort evaluations in this dataset, whereas overly quiet environments may correspond to greater thermal sensitivity. Accordingly, preserving moderate positive sound sources in quiet outdoor spaces may improve the overall environmental experience.

3.3.4 *Effects of acoustic comfort on thermal evaluation*

The results further indicate that acoustic comfort was associated with thermal evaluation. Under the same PET conditions, higher acoustic comfort was associated with lower thermal sensation votes, whereas the fitted slopes under different ACV categories were broadly similar (Figure 12(c)). This suggests that improved acoustic comfort may be associated with a more favorable thermal evaluation, even when the physical thermal conditions remain unchanged.

Figure 12(d) shows that the relationship between thermal comfort and PET also varied across acoustic comfort categories. The fitted vertex PETs under low, moderate, and high acoustic comfort conditions were 32.0 °C, 33.9 °C, and 37.2 °C, respectively, indicating that respondents in noisier environments tended to prefer relatively cooler thermal conditions. From a design perspective, this result suggests that in acoustically unfavorable outdoor spaces, shading, ventilation, and other heat-mitigation measures may be especially important for improving overall comfort.

3.3.5 *Effects of sound-source type on thermal evaluation*

The results indicate that perceived sound-source type was associated with thermal evaluation. The fitted slope between PET and thermal sensation was the lowest under negative sound-source conditions (k = 0.0423), whereas the slopes under neutral and positive sound-source conditions were higher and relatively similar (k = 0.09319 and 0.08619, respectively) (Figure 13(a)). When PET was below 30 °C, differences among sound-source types were limited. However, when PET exceeded 30 °C, the increase in thermal sensation with PET became flatter under negative sound-source conditions. The PET corresponding to TSV = 0 was 27.6 °C under negative sound-source conditions and 29.2 °C under positive sound-source conditions, suggesting that respondents exposed to less favorable acoustic conditions tended to prefer relatively cooler thermal environments.

Figure 13(b) further shows that sound-source type was associated with the relationship between thermal comfort and PET. Compared with negative sound-source conditions, positive sound-source conditions were associated with greater thermal acceptability and higher thermal tolerance under warmer conditions. These findings suggest that, in hot outdoor spaces, reducing intrusive noise and preserving or reinforcing positive sound sources may help improve thermal acceptability.

4 Discussion

4.1 Thermoacoustic environmental differences and thermal adaptability in historic spaces

This study showed that different types of historic spaces in Jinci Temple had distinct thermoacoustic characteristics. The ES site was an open hard-paved space. It had the highest mean T_a and G, at 36.7 °C and 419.4 W/m^2, respectively, and the lowest mean RH, at 30.3%. These results indicate the strongest summer heat exposure at ES. Its mean SPL was 71.5 dB, suggesting that intensive pedestrian activity in the entrance area also increased acoustic pressure. By contrast, the TS site was affected by canopy shading, and its mean G and T_a decreased to 137.3 W/m^2 and 34.5 °C, respectively. The WS site had the highest mean RH (44.1%) and the lowest mean SPL (61.3 dB). Although the SP site had the lowest mean G (3.5 W/m^2), it had the highest mean SPL (83.1 dB), which may be related to visitor gatherings and acoustic reflections within the semi-enclosed space.

The subjective evaluation results were generally consistent with the measured environmental conditions. The ES site showed relatively poor thermal comfort, with 68.2% of respondents reporting TCV ≤ −1. At the TS site, 91.7% of respondents reported TCV ≥ 0, indicating better thermal comfort in shaded spaces. For the acoustic environment, 97.3% of respondents at the WS site reported ACV ≥ 0, which was consistent with its lower SPL and natural sound-source characteristics. In contrast, the SP site received relatively lower acoustic comfort evaluations, further reflecting the adverse effects of visitor gatherings and acoustic reflections within the semi-enclosed space.

In addition, respondents in the summer survey at Jinci Temple showed relatively high thermal adaptability. The NPET was 29.4 °C, and the NPETR was 23.7–35.1 °C, both of which were higher than values reported in previous studies on other urban open spaces in cold regions, including parks, residential areas, and urban open spaces (Cohen et al. 2013; Mi et al. 2020; Yin et al. 2021; Liu et al. 2023a; Yuan et al. 2023; Zhen et al. 2025a) (Table 7). This suggests that thermal evaluation patterns in historic spaces may differ

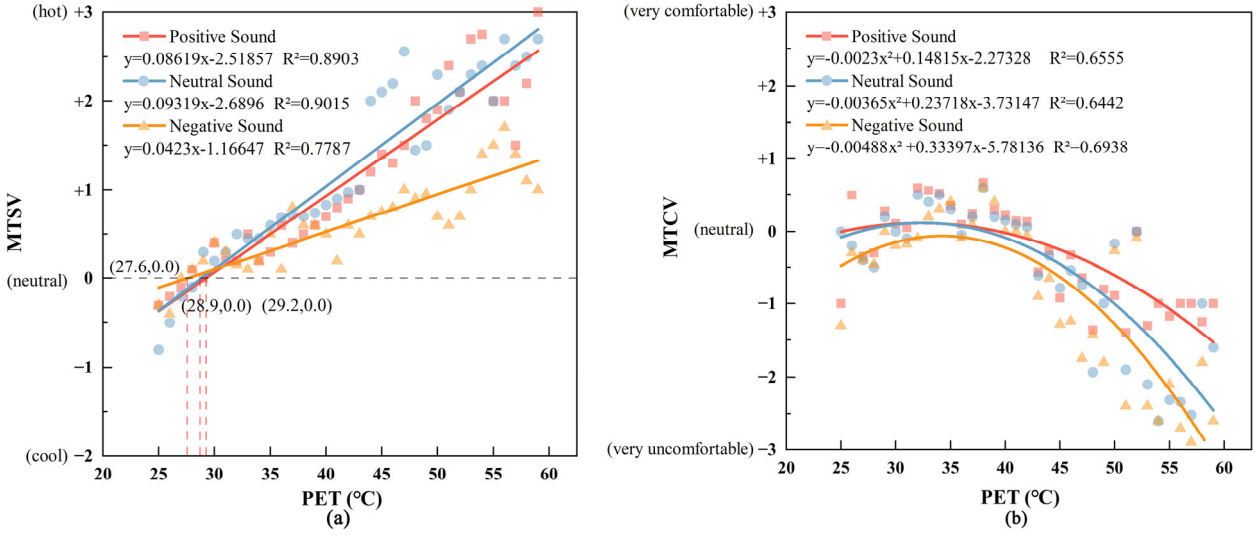


Fig. 13 Relationship between PET and thermal evaluation for different sound-source types: (a) PET and MTSV; (b) PET and MTCV

Table 7 Comparison of NPET and NPETR reported in different studies

City, country	Site	Visitor type	Season	NPET (°C)	NPETR (°C)
This study	Historic spaces	Residents and tourists	Summer	29.4	23.7–35.1
Harbin/China (Yin et al. 2021)	High-density business district	Resident	All seasons	—	13.0–20.0
Xi'an/China (Yuan et al. 2023)	Urban outdoor open spaces	Resident	All seasons	16.8	9.3–24.4
Tel Aviv/Israel (Cohen et al. 2013)	Urban park/town square	Resident	Summer	23.9	19.0–26.0
Xi'an/China (Liu et al. 2023a)	Riverfront landscape belt	Resident	All seasons	20.4	16.4–25.4
Xi'an/China (Mi et al. 2020)	Residential community	Resident	All seasons	19.7	12.4–26.9
Xi'an/China (Zhen et al. 2025a)	Ming City Wall Scenic Area	Resident	All seasons	17.1	13.5–20.7

from those in conventional urban open spaces. This difference may be related to the regulating effects of vegetation, water bodies, and heterogeneous underlying surfaces on local heat exposure (Sun et al. 2025). It may also be associated with visitors' low activity intensity, sightseeing and lingering behaviors, and specific place expectations (Li et al. 2024) in historic spaces. Because this study did not directly measure the relevant psychological variables, these explanations should be regarded as reasonable inferences from observational results rather than confirmed causal mechanisms.

4.2 Thermoacoustic interactions and explanatory mechanisms in historic spaces

This study found that the thermal and acoustic environments in historic spaces were not independent in their influence on comfort evaluation but were associated with each other. First, acoustic evaluation was influenced not only by SPL but also by sound-source type. When MASV = 0, the SPLs corresponding to positive, neutral, and negative sound sources were 72.2 dB, 68.8 dB, and 60.9 dB, respectively. When MACV = 0, the SPLs corresponding to the three sound-source types were 76.2 dB, 72.4 dB, and 66.2 dB, respectively. These results indicate that, at the same acoustic sensation or acoustic comfort level, respondents had higher tolerance for positive sound sources and were more sensitive to negative sound sources. Therefore, acoustic evaluation in historic spaces depends not only on SPL, but also on sound-source type and cultural context.

Second, thermal environmental conditions influenced acoustic evaluation. Under thermally comfortable conditions, the fitted slope between SPL and acoustic sensation was smaller, indicating that respondents were less sensitive to changes in SPL. Under thermally uncomfortable conditions, changes in the acoustic environment had a more pronounced effect on subjective evaluation. Meanwhile, sound-source type was also associated with thermal evaluation. Under higher PET conditions, respondents showed greater thermal acceptability in positive sound-source conditions. Specifically, the PET corresponding to MTSV = 0 was 27.6 °C under negative sound-source conditions and 29.2 °C under positive sound-source conditions. This suggests that positive sound sources may be associated with greater thermal acceptability under high-temperature conditions.

Based on these results, this study proposes an interpretive framework for understanding thermoacoustic interactions in historic spaces and provides an empirical explanation of the observed associations. First, at the physical environmental level, spatial morphology, paving materials, vegetation and water bodies, and visitor activity intensity jointly influence the thermal and acoustic environments. Second, at the multisensory perception level, the findings reveal bidirectional interactions between thermal sensation and acoustic sensation. Thermally comfortable conditions may reduce respondents' sensitivity to SPL variation, while quieter acoustic environments may enhance their perception of changes in the thermal environment. Third, at the psychobehavioral level, place meaning, visiting purpose, and behavioral patterns in historic spaces may increase visitors' tolerance of certain environmental stresses. Overall, outdoor comfort in historic spaces cannot be explained by a single thermal or acoustic indicator alone. Instead, physical environmental conditions, sound-source meanings, and visitor perception characteristics should be considered together.

4.3 Thermoacoustic environment optimization strategies based on simulation evaluation and HIA

This study employed ENVI-met (Liu et al. 2023b) and CadnaA (Zhou et al. 2024b) to quantify representative thermoacoustic environmental optimization scenarios. An exploratory HIA framework (Court et al. 2022) was subsequently applied to preliminarily compare the environmental benefits, heritage compatibility, and implementation feasibility of the proposed strategies.

4.3.1 Optimization scenarios and model validation

The optimization scenarios were developed based on field measurements, spatial morphology, vegetation structure,

parameter ranges reported in previous studies, and the software databases. ENVI-met (Ullah et al. 2024) was used to evaluate the thermal benefits of tree infill (Ji et al. 2025; Li et al. 2025b), green buffers, reversible shading (Martinez et al. 2025; Zeng et al. 2025), canopy-density regulation (Fu et al. 2025), and water–vegetation synergy (Cao et al. 2022), while CadnaA was used to assess the effects of green buffers on boundary noise propagation (Ow and Ghosh 2017). The baseline ENVI-met model was initialized using the measured meteorological data to define the initial and boundary conditions. The mean absolute errors (MAE) between the simulated and measured T_a, RH, V_a, and T_{mrt} were 0.86 °C, 4.2%, 0.12 m/s, and 4.35 °C, respectively. The CadnaA model was calibrated using equivalent sound sources derived from the observed sound-source distribution and measured SPL. The mean absolute difference between the simulated and measured SPLs was 1.2 dB. The models were primarily used to compare relative differences among the scenarios rather than to predict the absolute environmental improvements achievable after implementation.

Thermal performance was evaluated using ΔT_{mrt} (change in mean radiant temperature) and ΔPET (change in physiological equivalent temperature) (Lee et al. 2023), whereas acoustic performance was evaluated using ΔSPL (change in sound pressure level). All three indicators were defined as the baseline value minus the corresponding scenario value. The results were extracted at a height of 1.5 m, with the ENVI-met outputs obtained at 14:00 as a representative high-temperature time. Mean LAD in the scenarios represents the average leaf area density within the effective canopy volume; the detailed parameters are presented in Figures 14(a)–(d). All tree-substitution scenarios employed the same functional type of native deciduous broad-leaved trees, with the canopies assumed to be fully foliated under summer conditions. Because the representative tree type was held constant, differences among the simulated scenarios primarily reflected variations in crown width, planting spacing, and mean LAD rather than species-related differences.

4.3.2 Quantified scenarios and strategy pathways for different spaces

1) Vegetation shading and greening optimization

ES exhibited the most pronounced heat exposure, with mean T_a and G values of 36.7 °C and 419.4 W/m^2, respectively. Tree infill, multilayer green buffers, and reversible shading scenarios were therefore established for the boundary and central activity areas. The moderate and intensive tree-infill scenarios produced ΔT_{mrt} values of 4.6 °C and 5.8 °C and ΔPET values of 2.0 °C and 2.6 °C, respectively (Figure 14(a)). Given that the high-level scenario may increase visual obstruction, root-related impacts, and maintenance demands, moderate tree infill provides a more appropriate balance between thermal benefits and heritage conservation.

The 3-m and 5-m multilayer green buffers achieved ΔT_{mrt} values of 4.0 °C and 5.3 °C, ΔPET values of 1.7 °C and 2.3 °C, and ΔSPL values of only 0.2 dB and 0.4 dB, respectively. Because the predicted SPL reductions of 0.2–0.4 dB were smaller than the CadnaA calibration difference of 1.2 dB, they should not be interpreted as reliable evidence of substantial physical noise attenuation. The primary functions of the green buffers were therefore related to boundary shading and spatial-interface improvement. Multilayer green buffers are therefore not recommended as direct noise-control measures, but rather as integrated interventions offering shading, spatial-boundary enhancement, potential soundscape improvement, and heritage compatibility. Their value should not be assessed solely in terms of ΔSPL.

For the reversible shading scenarios with solar transmittances of 0.3 and 0.5, the corresponding ΔT_{mrt} values were 12.5 °C and 8.5 °C, while the ΔPET values were 5.3 °C and 3.6 °C, respectively. Among the simulated ES alternatives, the reversible shading scenario with a solar transmittance of 0.3 provided the greatest predicted thermal benefit. Given its predicted thermal benefit and reversible configuration, the HIA assessment identified this scenario as the preferred option for the central queuing area of ES.

The baseline mean LAD of TS was 0.24 m^2/m^3. Increasing mean LAD to 0.28 m^2/m^3 resulted in ΔT_{mrt} and ΔPET values of only 0.8 °C and 0.3 °C, respectively, whereas reducing mean LAD to 0.16 m^2/m^3 resulted in values of −1.5 °C and −0.6 °C, respectively (Figure 14(b)). Accordingly, maintaining the structure and health of the existing mature canopy and preventing foliage loss should be prioritized in TS, rather than introducing additional high-density planting.

Overall, lightweight reversible shading with a solar transmittance of 0.3 should be prioritized for the central queueing area of ES, while moderate tree infill and a 3 m multilayer green buffer may be applied along peripheral boundaries and transitional sections of visitor routes. For TS, priority should be given to maintaining the structure and health of the existing mature canopy.

2) Water–vegetation synergy and positive sound-source protection

WS had the lowest mean SPL, at 61.3 dB, and contained natural sound sources such as flowing water, birdsong, and wind-induced sounds. With the existing water conditions maintained, a 2-m-wide low vegetation belt and a 4-m-wide multilayer vegetation belt were simulated. The two scenarios produced ΔT_{mrt} values of 0.8 °C and 3.5 °C and ΔPET values of 0.3 °C and 1.5 °C, respectively (Figure 14(c)). Although the 4-m-wide multilayer vegetation belt produced

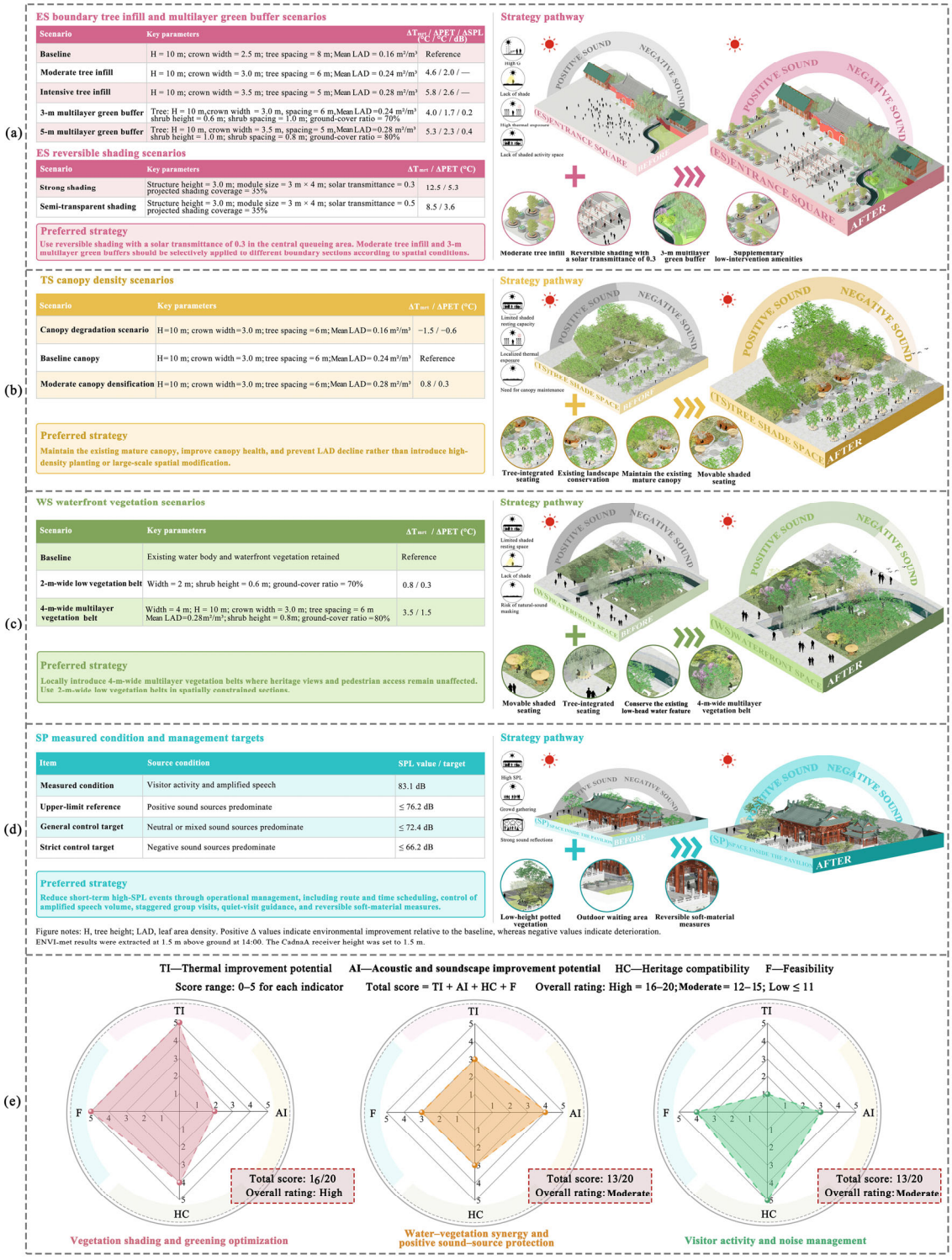


Fig. 14 Quantified thermoacoustic environmental optimization scenarios, strategy pathways, and HIA evaluation across different spaces: (a) ES; (b) TS; (c) WS; (d) SP; and (e) comprehensive HIA scores

greater predicted thermal benefits, it should be considered only in localized areas where it would not obstruct waterfront views, impede visitor circulation, or compromise the integrity of the historic water system. In more spatially constrained areas, the 2-m-wide low vegetation belt is preferable because of its more limited visual and spatial intervention. No separate CadnaA scenario was developed for WS because the strategy focused on preserving positive natural sounds rather than physical noise reduction. The acoustic strategy for WS should focus on preserving the

audibility of flowing water, birdsong, and wind-induced sounds while reducing their masking by mechanical noise, public-address systems, and high-level amplified speech.

3) Visitor activity and noise management

SP had a mean SPL of 83.1 dB and exhibited pronounced short-term fluctuations. Because the locations and sound power levels of visitor conversations, tour-guide commentary, and amplification devices varied substantially, no stable CadnaA prediction scenario was established. Perception-based management reference values were derived from the relationships between the SPLs averaged during questionnaire completion and the corresponding acoustic comfort evaluations (Figure 14(d)). Management should apply tiered control according to the dominant perceived sound-source type. The following values should be interpreted as perception-based management references for the same SPL averaging period used during questionnaire completion, rather than as instantaneous or regulatory noise limits. When positive sound sources predominate, 76.2 dB may be used as an upper perception-based management reference value. When neutral or mixed sound sources predominate, 72.4 dB may be used as the corresponding perception-based management reference value. When negative sound sources predominate, a stricter perception-based management reference value of 66.2 dB may be applied. Recommended measures include limiting the volume of amplification devices, optimizing tour-guide routes and schedules, avoiding the simultaneous congregation of multiple groups, reducing short-term crowding, and implementing time-staggered visitation. Where necessary, reversible measures such as movable upholstered seating, removable fabric acoustic screens, or low-height potted plants may be introduced, provided that they neither obstruct historic architectural elements nor alter the significant spatial character of the pavilion.

4.3.3 Exploratory HIA-based integrated evaluation and strategy prioritization

Building on the scenario simulations, this study applied an exploratory HIA framework to compare three representative strategy categories. The candidate strategies were assessed across four dimensions: thermal improvement potential (TI), referring to the capacity to improve thermal comfort; acoustic environment and soundscape improvement potential (AI), referring to the capacity to reduce intrusive noise or increase the proportion of positive sound sources; heritage compatibility (HC), referring to the extent to which the interventions are consistent with heritage values, historic character, spatial configuration, and site integrity; and feasibility (*F*), referring to technical implementability, relative implementation burden, and maintenance requirements. Each indicator was scored on a scale from 0 to 5, with higher scores indicating greater improvement potential, stronger heritage compatibility, or higher implementation feasibility. The total score was calculated as TI + AI + HC + *F*, with a maximum of 20 points. For this exploratory comparison, the authors defined three overall score ranges: high (16–20), moderate (12–15), and low (≤11) (Figure 14(e)).

Three members of the author team with different research backgrounds independently assigned scores based on the field measurements, simulation results, fitted relationships, site observations, and heritage conservation constraints. The final scores were determined through discussion and consensus. Equal weighting was adopted in the preliminary comparison of the strategies to balance thermal benefits, acoustic benefits, heritage conservation, and implementation feasibility. Because the assessment was intended as an exploratory comparison rather than a formal decision-analysis procedure, inter-rater reliability statistics were not calculated.

Under this evaluation framework, vegetation shading and greening optimization achieved a total score of 16 and was classified as "high." This indicates substantial thermal improvement potential and strong heritage compatibility, particularly when implemented through localized, reversible, and moderate spatial interventions. Reversible shading was predicted to improve highly exposed activity areas, while moderate tree infill and a 3-m multilayer green buffer could improve the boundary microenvironment with relatively limited spatial intervention. Canopy maintenance in TS can also prevent deterioration of the existing thermal advantages with minimal intervention.

The combined strategy of water–vegetation synergy and positive sound-source protection received a total score of 13 and was classified as "moderate." This strategy can improve the local thermal environment and preserve natural soundscapes under specific spatial conditions. However, its applicability is constrained by waterfront dimensions, historic sightlines, visitor circulation, and the landscape integrity of the historic water system. It is therefore more suitable for localized and phased implementation.

The combined visitor activity and noise management strategy also received a total score of 13 and was classified as "moderate." These measures involve limited physical intervention and exhibit high heritage compatibility; however, their effectiveness depends strongly on sustained operational management and visitor behavior, making stable prediction of physical improvement difficult at present. Visitor activity and noise management are therefore more appropriately implemented as low-intervention supplementary measures integrated into routine site operations.

Taken together, the simulation and HIA results indicate that vegetation shading and greening optimization should be

prioritized, particularly through reversible shading, moderate tree infill, and mature-canopy maintenance. Water–vegetation synergy and positive sound-source protection may be implemented locally according to waterfront spatial conditions, while visitor activity and noise management should serve as a continuous operational support strategy.

5 Conclusion

This study investigated outdoor comfort under thermoacoustic interactions at Jinci Temple, a representative historic space in China's cold region.

(1) The four representative open-space types showed distinct thermoacoustic environmental characteristics, and respondents exhibited relatively high thermal adaptability under hot summer visiting conditions. ES, an open hard-paved plaza, showed the strongest heat exposure. TS presented intermediate thermoacoustic conditions under semi-shaded tree-canopy cover. SP, a building-enclosed semi-open space, had the lowest mean *G* but the highest mean SPL. WS, a waterfront resting space, had the highest mean RH and the lowest mean SPL. During the summer survey period, the NPET was 29.4 °C, the NPETR was 23.7–35.1 °C, and the upper acceptable PET threshold was 34.3 °C. These findings suggest that thermal evaluation in historic spaces may differ from that in conventional urban open spaces.

(2) Thermal and acoustic perceptions were mutually associated, and sound-source type played an important role in thermoacoustic evaluation. Under thermally comfortable conditions, respondents were less sensitive to changes in SPL. In relatively quiet acoustic environments, respondents were more sensitive to changes in PET. When MASV = 0, the corresponding SPLs for positive, neutral, and negative sound sources were 72.2 dB, 68.8 dB, and 60.9 dB, respectively. This indicates that positive sound sources were associated with higher acoustic tolerance. Sound-source type was also associated with thermal evaluation. The neutral PET under negative sound-source conditions was 27.6 °C, which was lower than the value of 29.2 °C under positive sound-source conditions.

(3) Based on the field measurements, this study proposes an interpretive framework for thermoacoustic associations in historic spaces, involving the combined influences of the physical environment, multisensory perceptual modulation, and psychological and behavioral adaptation. Among the simulated alternatives, reversible shading with a solar transmittance of 0.3 produced the greatest predicted local thermal benefits, with ΔT_{mrt} and ΔPET values of 12.5 °C and 5.3 °C, respectively. The exploratory HIA evaluation ranked vegetation shading and greening optimization as high, with a total score of 16/20. Water–vegetation synergy and positive sound-source protection and visitor activity and noise management each received a score of 13/20 and were ranked as moderate. Accordingly, priority should be given to reversible shading, moderate tree infill, and maintenance of existing mature canopies, whereas waterfront vegetation optimization should be implemented conditionally and visitor noise management should serve as a supplementary operational measure.

The findings of this study mainly reflect typical hot summer visiting conditions and may provide a reference for historic temples, historic gardens, and heritage tourism spaces with similar spatial characteristics. Given that season, climatic region, spatial type, and visitor activity patterns may influence thermoacoustic comfort evaluations, the proposed thresholds and optimization strategies should be calibrated against local field measurements before being applied to other contexts. The thermoacoustic associations and interpretive framework derived from this single case study should be regarded as empirical insights rather than a universally applicable causal model. The ENVI-met and CadnaA simulations were primarily used to compare the relative improvement trends among different scenarios; therefore, the results should not be interpreted as absolute improvements achievable after engineering implementation or as universally optimal parameter values. Future research should incorporate long-term, multi-seasonal field monitoring across multiple case-study sites, together with independent model validation, sensitivity analysis, multivariate modelling, and coupled thermal–acoustic simulations, to further examine the environmental effects, robustness, and applicability of different combinations of tree species, mean LAD, planting spacing, and canopy structure.

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Declaration of competing interest

The authors have no competing interests to declare that are relevant to the content of this article.

Author contribution statement

Rong Zhang drafted the manuscript. Lingling Cui designed the experiment and revised the manuscript. Jingjing Liu conducted the questionnaire survey. Weijie Xia provided some translations. Zijie Wang and Meng Zhen provided resources and edited the manuscript.

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